Lost in Translation: Measuring English-Lens Bias in a Multilingual LLM's Answers About Polish Culture

Are Claude's answers about culturally significant Polish topics changing with the language of the question?

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Abstract

Large language models are trained mostly on English text, which raises the concern that their knowledge of non-English cultures is filtered through an Anglophone lens even when they answer in the native language. This project tests that concern on one model (Claude Sonnet 4.5) and one language pair (English–Polish). Twenty-five translation-matched prompts about culturally significant Polish topics (contested history, literature, and social norms) were posed in both languages under identical generation settings, with matched English and Polish Wikipedia articles serving as a human-discourse baseline. English and Polish answers to the same question were semantic near-duplicates (median cross-lingual cosine similarity .81), yet their affective framing diverged: Polish answers were significantly less negative than English ones (Wilcoxon signed-rank test, $p = .0004$, rank-biserial $r = +.76$), with the gap concentrated in the contested-history domain. A "native pull" index of anchoring to same-language Wikipedia trended in the predicted direction, Polish answers less anchored in Polish discourse, but did not reach significance ($p = .53$, $n = 24$). Together, the results suggest the model serves a single, largely language-independent content core through language-specific emotional registers.

1 Background

The web-scale corpora that large language models learn from are predominantly English; mid-resource languages such as Polish contribute only a small fraction of the training tokens (Joshi et al., 2020; Bender et al., 2021). This imbalance is usually discussed as a performance problem as the models are simply worse in smaller languages, but it has a subtler representational consequence: most of what a model "knows" about Poland was written in English, by and for an international audience.

That is the concern this project calls the "English lens." When a user asks a model about the Warsaw Uprising in Polish, fluent Polish on the surface does not guarantee that the answer is grounded in Polish-language source. Comprehensibility work suggests that multilingual transformers route non-English inputs through internal representations closest to English before producing output in the target language (Wendler et al., 2024), and cultural-bias studies find that models default to Anglo-American norms and framings even when prompted in other languages (Cao et al., 2023; Naous et al., 2024). If so, answers in Polish may be, in effect, translations of an English-shaped answer.

Polish is a good test case: a mid-resource language with a large, self-contained discourse tradition, and a set of topics on which Polish-language and English-language framings genuinely diverge. Contested national history: Katyń, the 1944 Warsaw Uprising, Jedwabne, the Volhynia

massacres, the Smolensk disaster — carries a commemorative, often martyrological register in Polish public memory that external English-language accounts typically render in a cooler, more critical or tragedy-centred tone. Literature and everyday social customs provide two comparison domains where the affective stakes are progressively lower.

Research question. Does English-language dominance in LLM training data cause a model to reframe culturally significant Polish topics through an Anglophone lens, even when responding in Polish? The question is operationalized as three hypotheses. **H1 (homogenization):** English and Polish answers to the same prompt will be semantically near-identical (high cross-lingual cosine similarity), consistent with both surface forms being generated from one shared underlying representation. **H2 (anchoring):** Polish answers will be less anchored in Polish-language discourse (Polish Wikipedia) than English answers are in English-language discourse: native pull(PL) < native pull(EN). **H3 (affective framing):** the sentiment of paired answers will diverge by prompt language, most strongly in the contested-history domain.

2 Method

2.1 Design

The study uses a within-topic paired design. The independent variable is the **language of the prompt** (English vs Polish): each of 25 culturally significant topics was posed as a translation-matched question pair — identical content, different language — across three domains: contested history (9 topics), literature (8), and social norms (8). Controlled variables: a single fixed model, temperature 0 (deterministic decoding), an identical maximum-token cap (600) for every call, matched prompt wording and register, and identical „ ” quotation marks in both language versions of every prompt. The dependent variables are cross-lingual semantic similarity, anchoring to same-language Wikipedia (“native pull”), and sentiment polarity; response length in words serves as a control measure.

2.2 Materials

Prompts were generated programmatically (*kasia_code.py build*) into *prompts.csv*; each row holds a prompt ID, domain, topic, the English and Polish questions, and the titles of the matched English and Polish Wikipedia articles. Example pair (h03): “What was the Katyn massacre and why is it significant in Polish history?” / „Czym była zbrodnia katyńska i dlaczego jest ważna w historii Polski?”.

2.3 Model responses

Responses were collected through the Anthropic API (*kasia_code.py collect*) from *claude-sonnet-4-5* in June 2026 — one response per prompt × language (N = 50) — with temperature 0, max_tokens = 600, exponential-backoff retries (up to 4 attempts, 5–20 s), and resumable collection so that interrupted runs never duplicate rows. Each response was stored in *data/llm_responses.csv* together with word and character counts and a timestamp. One reproducibility caveat: *claude-sonnet-4-5* is an alias that resolves to the provider's latest dated snapshot of that model, so exact regeneration requires pinning the dated snapshot current at collection time.

2.4 Human baseline (data acquisition)

For each topic, the matched English and Polish Wikipedia articles were retrieved programmatically (*kasia_code.py wiki*) from the MediaWiki API using the *wikipedia-api* Python

library with a declared academic User-Agent, giving 48 articles (24 topics × 2 languages). One topic (s06, the untranslatable concept *kombinować*) has no matching Wikipedia article in either language, so no titles were assigned in *prompts.csv*; it was logged to *wiki_missing.csv* and excluded from baseline-dependent analyses. Both the article summary (lead section) and the full text were stored; the analyses use the summary, because full articles running to tens of thousands of words cannot be meaningfully compared, as a single embedding, against ~200-word answers.

2.5 Measures

All texts were embedded with **LaBSE** (Feng et al., 2022), a sentence encoder trained specifically to place translation pairs close together across 109 languages, which suits an English–Polish comparison; vectors were L2-normalized so cosine similarity equals the dot product. Three indices were computed per topic. (1) **Homogenization** = cosine(EN answer, PL answer). (2) **Native pull**, computed separately for each answer as cosine(answer, same-language Wikipedia summary) - cosine(answer, other-language Wikipedia summary); the English-lens prediction is pull(PL) < pull(EN). (3) **Sentiment polarity** = P(positive) - P(negative) from the multilingual XLM-RoBERTa sentiment classifier *cardiffnlp/twitter-xlm-roberta-base-sentiment* (Barbieri et al., 2022), applied identically to English and Polish responses with truncation at 512 tokens.

2.6 Statistical analysis

Because the samples are small (n = 25 topics; n = 24 for baseline-dependent measures) and no distributional assumptions were warranted, paired hypotheses were tested with two-sided Wilcoxon signed-rank tests (Wilcoxon, 1945); medians are reported throughout, and the matched-pairs rank-biserial correlation is reported as the effect size. The pipeline is implemented in Python (pandas, NumPy, SciPy, sentence-transformers, transformers, Matplotlib, seaborn) in two scripts: *kasia_code.py* (prompt building, response collection, Wikipedia retrieval) and *kasia_code_analyze.py* (metrics, statistics, figures).

2.7 Deviations from the proposal

Three deliberate changes were made relative to the project proposal. First, the study uses one model (Claude) rather than Claude and Gemini: a single-model, within-topic paired design keeps every comparison controlled and the workload feasible, at the cost of generality (4.4). Second, LaBSE replaced *paraphrase-multilingual-mpnet-base-v2* as the embedding model, being purpose-built for cross-lingual translation-pair similarity. Third, the proposal's corpus-level “bias gap” was sharpened into the per-answer native-pull index (2.5), which tests anchoring topic-by-topic and supports a paired statistical test.

3 Results

3.1 Homogenization (H1)

Across the 25 topics, the median cosine similarity between the English and the Polish answer to the same question was **.81** (range ≈ .76–.87): the model delivers close to the same content whichever language it is asked in (Figure 1). Similarity was highest and tightest in contested history (median ≈ .82) and lowest in social norms (≈ .78), where culture-specific customs leave more room for language-specific detail. The values sit somewhat below the ≥ .85 anticipated in the proposal, so H1 receives qualified support: strong homogenization, though not total identity.

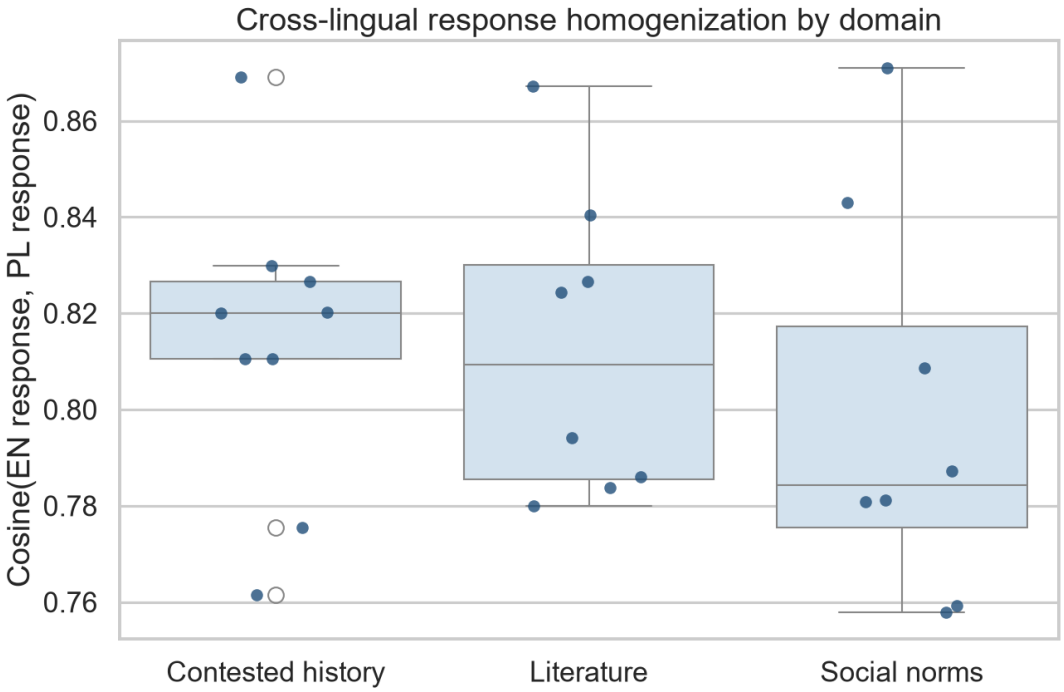


Figure 1. Cosine similarity between the English and Polish answers to the same prompt, per domain (each point = one topic, n = 25; boxes show the median and interquartile range). Higher values mean the model delivers more nearly identical content regardless of question language. Overall median = .81.

3.2 Affective framing (H3) — the main finding

Same content, different tone. Polish answers were systematically **less negative** than their English twins: median polarity −.10 (PL) vs −.19 (EN); Wilcoxon signed-rank test, n = 25, **p = .0004**, rank-biserial **r = +.76** - a large effect. In Figure 2 the large majority of topics lie above the EN = PL diagonal. The divergence is concentrated exactly where H3 predicts: the median Polish-minus-English difference was largest in contested history (+.20), smaller in literature, and near zero in social norms. In plain terms, asked in English about Katyń, Volhynia or Jedwabne, the model adopts a distinctly darker, more negative register than when asked the same question in Polish.

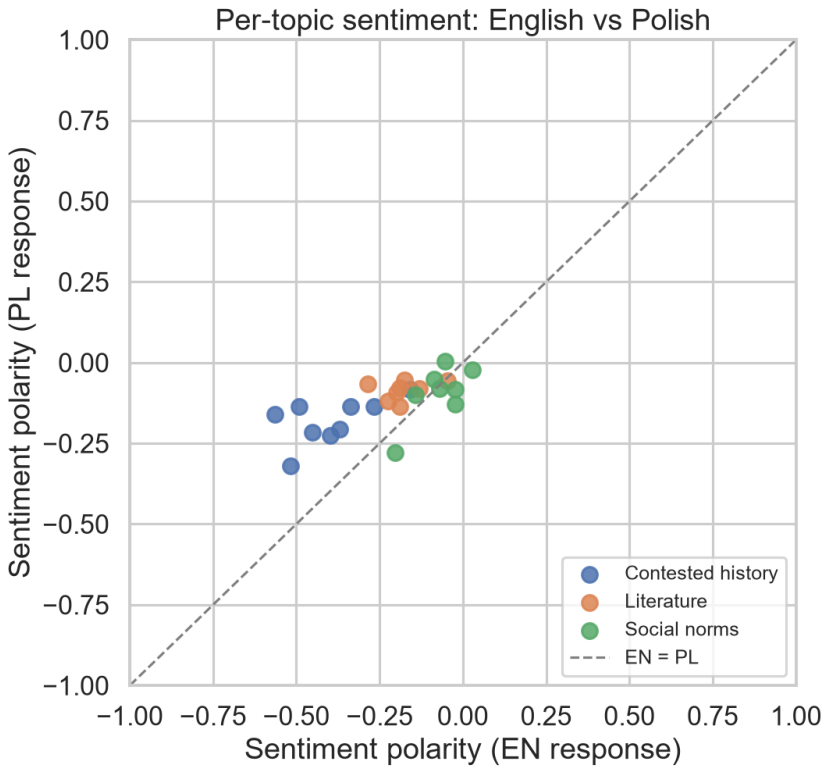


Figure 2. Sentiment polarity (P(pos) - P(neg)) of each topic's English vs Polish answer, coloured by domain. Points above the dashed EN = PL line are more positive in Polish; below, more positive in English. Polish answers are significantly less negative overall (Wilcoxon signed-rank test, n = 25, p = .0004, rank-biserial r = +.76), with the largest gaps in contested history (median PL − EN difference +.20).

3.3 Anchoring to native discourse (H2)

Both English and Polish answers sat closer to the Wikipedia article in their own language than to the other-language article (both median pulls positive), and the pattern pointed in the predicted direction: Polish answers were less anchored in Polish Wikipedia (median native pull +.067) than English answers were in English Wikipedia (+.091). The paired difference, however, was not statistically significant (Wilcoxon signed-rank test, n = 24, **p = .53**). H2 is therefore not supported at conventional thresholds: the direction is consistent with the hypothesis, but with this sample the effect cannot be distinguished from chance (Figure 3).

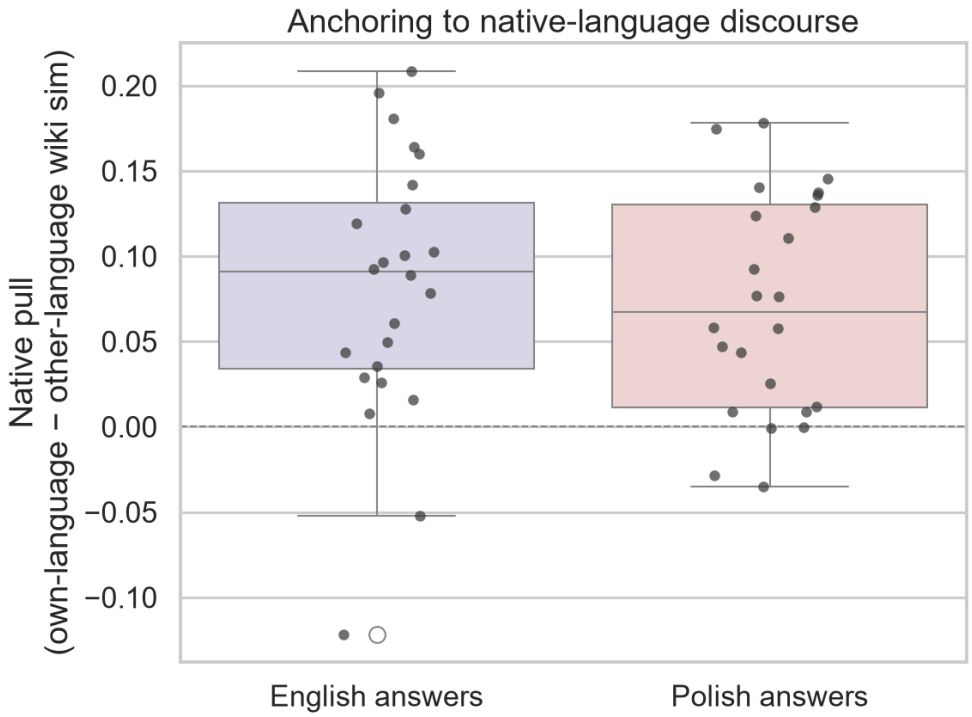


Figure 3. "Native pull" = how much closer an answer is to the Wikipedia article in its *own* language than to the article in the other language (n = 24; topic s06 has no baseline). If Polish answers show less native pull than English answers, Polish output is comparatively less anchored in Polish discourse - the English-lens signal. The medians trend in that direction (+.067 vs +.091), but the paired difference is not significant (Wilcoxon, p = .53).

3.4 Length control

All calls shared the same max-token cap, yet Polish responses came out about a fifth shorter in words (median 165 vs 211; Figure 4). This is a tokenization artefact, Polish words cost more tokens, so the cap equalizes token budgets not word counts. The asymmetry is consistent across domains and is treated as a limitation (§4.4) rather than a confound: the similarity and sentiment measures operate on embedded or classified content, not on raw length.

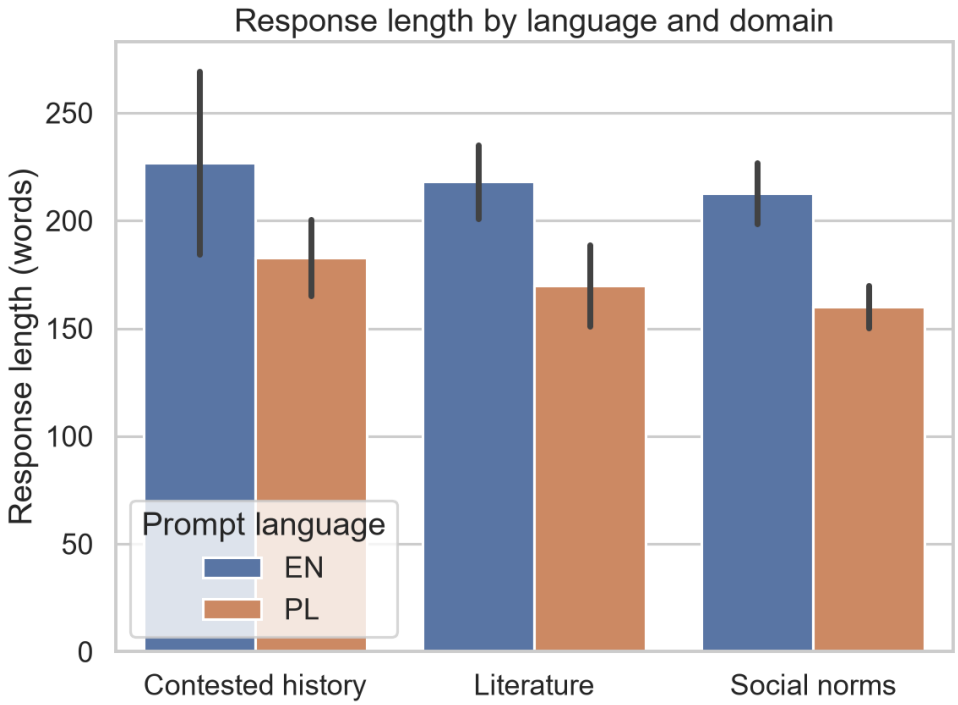


Figure 4. Mean response length in words by domain and prompt language; error bars = SD. Control check on the matched max-token cap: token budgets were identical, yet Polish responses are ≈20% shorter in words (median 165 vs 211) because Polish is more token-dense, a limitation discussed in 4.4.

4 Discussion

4.1 Summary of findings

The three results form one coherent picture. The model gives largely the same answer in both languages (median cosine .81) and shows no significant difference in how strongly each language's answers anchor to their own Wikipedia discourse, but it delivers that shared content in language-specific emotional registers, with English answers about Polish historical traumas markedly more negative than Polish ones. The strongest, statistically robust language effect in this study is therefore not *what* the model says, but the tone in which it says it.

4.2 Does this support the English-lens hypothesis?

Partly, and in an informative way. In Figure 2 the topic-to-topic structure of sentiment is preserved across languages - points shift roughly parallel to the diagonal rather than scattering - which fits the view that a single underlying stance toward each topic, formed in a predominantly English-shaped representation space (Wendler et al., 2024), is re-rendered per language, with Polish receiving a milder register. Read this way, the sentiment gap is the English lens made visible: the cooler, externally framed English account is the default, and the Polish surface is an affective localization of it. The honest alternative reading is genuine adaptation: the model may be tracking real register differences between Anglophone and Polish discourse about these events, since Polish commemorative discourse frames the Uprising or Katyń through heroism and martyrdom, not only catastrophe. The present design cannot fully separate the two readings; the decisive comparison, the sentiment of the Wikipedia baselines themselves in both languages, i.e. the four-condition design sketched in the proposal, is the natural next step. What the data do rule out is full homogenization of affect: prompt language demonstrably changes framing even

when content is nearly constant, which is precisely the kind of subtle, user-invisible divergence the English-lens debate is about.

4.3 Interpreting the null result on native pull

The anchoring index moved in the predicted direction but fell far short of significance. Three non-exclusive explanations. *Power:* n = 24 topics can only detect fairly large paired effects. *Instrument:* LaBSE is trained to minimize the distance between translation equivalents - the very property that makes it good for cross-lingual matching may compress the language-specific differences that native pull tries to measure. *Baseline genre:* Wikipedia lead sections are deliberately neutral and encyclopedic in both languages, so the English and Polish baselines may themselves be too similar for differential anchoring to show. The result is reported as a genuine null with a directional trend, not as evidence that anchoring differences are absent.

4.4 Limitations

The study covers a single model and a single language pair, so generalization is untested. Each cell contains one deterministic response (temperature 0), trading sampling noise for a single point estimate of the model's default answer. The sentiment classifier is tuned on tweets, not encyclopedic prose, and truncates input at 512 tokens, so polarity should be read as a coarse signal; moreover, its cross-lingual calibration cannot be verified here - if the classifier were systematically harsher on English than on Polish text, part of the sentiment gap could be instrumental. The domain-specificity of the effect argues against a purely instrumental explanation (a calibration artefact should produce a roughly constant gap, not +.20 in contested history and ≈0 in social norms), but does not eliminate it. Wikipedia is an imperfect proxy for “native discourse”. Finally, the matched token cap does not equalize word counts across languages (§3.4), and the model alias means the underlying snapshot may drift over time.

4.5 Conclusion

Millions of people ask language models about their own history in their own language. For the model and topics studied here, the answer to the research question is: mostly the same facts, but not the same feeling , and the gap opens widest exactly on the topics that matter most to Polish collective memory.

5 Resources

5.1 References

Barbieri, F., Espinosa Anke, L., & Camacho-Collados, J. (2022). XLM-T: Multilingual language models in Twitter for sentiment analysis and beyond. *Proceedings of LREC 2022*, 258–266.

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5.2 Software and data sources

Anthropic Claude API (model *claude-sonnet-4-5*; responses collected June 2026); *Wikipedia-API* Python library over the MediaWiki API; English and Polish Wikipedia (text licensed CC BY-SA 4.0); *sentence-transformers* with LaBSE; Hugging Face *Transformers* with *cardiffnlp/twitter-xlm-roberta-base-sentiment*; pandas, NumPy, SciPy, Matplotlib, and seaborn under Python 3.

5.3 Project materials

All materials accompany this report in the submission ZIP: *kasia_code.py* (prompt building, response collection, Wikipedia retrieval), *kasia_code_analyze.py* (metrics, statistics, figures), and the *kasia_data* folder (*prompts.csv*, *llm_responses.csv*, *wiki_articles.csv*, *wiki_missing.csv*, *analysis_per_prompt.csv*, *stats_summary.txt*), together with the four figures reproduced above.